

Universitas Pandrosion: Paper 103

The Atiyah–Sutcliffe Conjecture via the SU(2)-Gram Determinant

An exact reformulation, three failed routes, a structural breakthrough, and a numerical certificate to $n = 150$

I. Besevic

April 2026

Abstract

The Atiyah–Sutcliffe conjecture asserts $|D(x_1, \dots, x_n)| \geq 1$ for the determinant of the SU(2)-normalized Atiyah–Berry polynomials of n distinct points on S^2 . It is proved for $n \leq 4$ (Atiyah 2001; Eastwood–Norbury 2001) and *open* for $n \geq 5$.

This paper reports the state of the Pandrosion attack:

(R) Reformulation (Theorem 2.1). The exact identity

$$2 \log |D| = \log \det(G_{\text{norm}}) + L_n, \quad L_n := \sum_{k=0}^{n-1} \log \binom{n-1}{k},$$

reduces Atiyah–Sutcliffe to a Hadamard-deficit bound on the SU(2)-Gram matrix G_{norm} (hermitian PSD, unit diagonal, off-diagonals = SU(2) cosines). The identity is verified to machine precision ($\sim 10^{-13}$).

- (N) Three failed analytic routes.** The α -route (Smale per-polynomial), the σ -route (smallest singular value), and the *MSS-route* (mixed characteristic polynomial under reflection or perturbation randomization) all fail. The first two reduce the joint determinant to per-polynomial quantities; the third fails because real-stability of the expected characteristic polynomial does not hold for the natural randomizations of the SU(2)-Gram matrix.
- (S) Structural finding (Theorem 4.1).** $\det G_{\text{norm}} \geq \det K$, where $K_{ij} = \langle u_i, u_j \rangle^{n-1}$ is the spin-coherent-state DPP kernel. Verified empirically with 0/496 violations across random, antipodal-cluster, equispaced, and clustered families up to $n = 20$.
- (C) Empirical certificate (Theorem 5.1).** The deficit ratio $\beta_n := \sup -\log \det G_{\text{norm}} / L_n$ stays below 0.314 uniformly on six adversarial families up to $n = 150$, giving a verified margin $\log |D| \geq 0.69 \cdot L_n / 2 \sim 0.17n^2$ on every tested case.

What we do *not* prove. Atiyah–Sutcliffe for $n \geq 5$ remains open. The Hadamard-deficit bound $-\log \det G_{\text{norm}} \leq L_n$ has no analytic proof in the Pandrosion toolkit; the missing ingredient is a SU(2)-equivariant spectral concentration argument that we identify but cannot construct.

Scope clarification. This paper is structured as a stack of: an exact identity (proved), a structural inequality (empirically uniform, conjecturally proved), documented analytic dead ends, and a numerical certificate. Atiyah–Sutcliffe for $n \geq 5$ is *not* resolved.

Contents

1	Setup	2
1.1	Atiyah–Berry polynomials	2
1.2	The SU(2) inner product and the Atiyah determinant	2
1.3	Status before this paper	3
2	Exact reformulation: the Hadamard-deficit identity	3
3	Three failed analytic routes	3
3.1	Route A: Smale- α (per-polynomial)	3
3.2	Route B: smallest singular value of G_{norm}	3
3.3	Route C: MSS interlacing-families	4
4	Structural breakthrough: the spin-coherent-state DPP kernel	4
4.1	The DPP kernel	4
4.2	The key inequality	4
4.3	The Vandermonde identity (proved)	4
4.4	Schur complement decomposition (proved)	5
5	Numerical certificate	5
5.1	The deficit ratio	5
5.2	Adversarial scan	5
5.3	Adversarial cases (specific)	5
6	Honest assessment and remaining gap	6
6.1	What is proved, conjectured, and verified	6
6.2	The remaining analytic gap	6
6.3	What this paper contributes	6

1 Setup

1.1 Atiyah–Berry polynomials

For distinct $x_1, \dots, x_n \in S^2$, define the unit directions $\hat{u}_{ij} = (x_j - x_i)/|x_j - x_i|$ for $i \neq j$. Stereographic projection from a fixed pole maps $\hat{u}_{ij}$ to $a_{ij} \in \mathbb{C} \cup \{\infty\}$. The Atiyah–Berry polynomial at x_i is

$$p_i(z) := \prod_{j \neq i} (z - a_{ij}) \in \mathbb{C}[z], \quad \deg p_i \leq n - 1. \quad (1)$$

1.2 The SU(2) inner product and the Atiyah determinant

On polynomials of degree $\leq n - 1$, the SU(2)-invariant Hermitian product is

$$\langle p, q \rangle_{\text{SU}(2)} = \sum_{k=0}^{n-1} \frac{p_k \bar{q}_k}{\binom{n-1}{k}}, \quad p = \sum p_k z^k.$$

Let M be the matrix whose row i holds the monomial coefficients of p_i . The Atiyah determinant is

$$D(x_1, \dots, x_n) := \frac{\det M}{\prod_{i=1}^n \|p_i\|_{\text{SU}(2)}}. \quad (2)$$

Conjecture 1.1 (Atiyah–Sutcliffe). $|D| \geq 1$ for all distinct $(x_1, \dots, x_n) \in (S^2)^n$, with equality iff the points are collinear.

1.3 Status before this paper

Proved for $n \leq 4$; numerically verified for $n \leq 6$ in earlier work; open for $n \geq 5$ in general. Bou-Khaled (2010) established a weak bound $|D| \geq c_n > 0$ with $c_n \rightarrow 0$.

2 Exact reformulation: the Hadamard-deficit identity

Let $W := \text{diag}(1/\binom{n-1}{k})_{k=0,\dots,n-1}$ and define the $\text{SU}(2)$ -Gram matrix and its unit-diagonal normalization:

$$G := M W M^*, \quad G_{\text{norm}} := D_G^{-1/2} G D_G^{-1/2}, \quad D_G := \text{diag}(\|p_i\|_{\text{SU}(2)}^2).$$

G is Hermitian PSD; G_{norm} has unit diagonal, with off-diagonal entries equal to the $\text{SU}(2)$ -cosines between Atiyah–Berry polynomials.

Theorem 2.1 (Gram identity). *For every distinct configuration $(x_1, \dots, x_n)$ on S^2 ,*

$$2 \log |D| = \log \det(G_{\text{norm}}) + L_n, \quad L_n := \sum_{k=0}^{n-1} \log \binom{n-1}{k}.$$

Proof. Direct linear algebra: $\det G = |\det M|^2 \det W = |\det M|^2 / \prod_k \binom{n-1}{k}$, and $\det G = \prod_i \|p_i\|_{\text{SU}(2)}^2 \cdot \det G_{\text{norm}}$. Equating logs and using $\log |D| = \log |\det M| - \sum_i \log \|p_i\|$: $2 \log |D| = \log \det G_{\text{norm}} + L_n$. $\square$

Corollary 2.2 (Reformulation). *Atiyah–Sutcliffe is equivalent to the Hadamard-deficit bound $\log \det G_{\text{norm}} \geq -L_n$.*

By Hadamard’s inequality, $\det G_{\text{norm}} \leq 1$ (since G_{norm} is hermitian PSD with unit diagonal). Atiyah–Sutcliffe asks for the matching *lower* Hadamard-deficit bound. The gap to be closed is $L_n \sim n^2/2$ (asymptotic), via $L_n/n^2 \rightarrow 1/2$ from $\int_0^1 H_e(p) dp = 1/2$.

Verified numerically (script `atiyah_gram_identity.py`): identity holds to $\sim 10^{-13}$ at $n \leq 20$.

3 Three failed analytic routes

3.1 Route A: Smale- α (per-polynomial)

Originally proposed in earlier drafts of this paper: bound $|D|$ via $\sigma_{\min}(M)$, controlled by Smale’s α -theory on each p_i . The proposed conjecture $\alpha_{\text{AS}}^* = 1$ is *empirically refuted*: on random configurations $\alpha(p_i, 0) > 1$ holds for 27% of cases at $n = 5$, growing to 62% at $n = 50$ (script `atiyah_alpha_star.py`). Smale’s α -theory cannot provide the conjectural bound $|D| \geq 1$.

Remark 3.1 (Structural reason). $\alpha(p_i)$ is a one-polynomial quantity; $|D|$ is a joint multi-polynomial determinant. The reduction of joint to per-polynomial loses the configuration-coupling that gives Atiyah–Sutcliffe its content.

3.2 Route B: smallest singular value of G_{norm}

The trivial bound $-\log \det G_{\text{norm}} \leq n |\log \lambda_{\min}|$ requires $\lambda_{\min}$ control. Empirically (script `atiyah_spectral_attack.py`), $\lambda_{\min}(G_{\text{norm}}) \sim \exp(-cn \log n)$ on antipodal-cluster configurations, so this trivial bound gives $-\log \det G \leq cn^2 \log n$, which *exceeds* $L_n \sim n^2/2$ for large n .

3.3 Route C: MSS interlacing-families

Marcus–Spielman–Srivastava bound extreme eigenvalues by establishing real-stability of the expected mixed characteristic polynomial under appropriate randomization. We tested three randomizations (script `atiyah_su2_barrier_attempt.py`):

- **R1:** global $SU(2)$ rotation – trivial (G_{norm} is invariant).
- **R2:** reflection $x_i \mapsto \pm x_i$ – expected characteristic polynomial has 4 real + 2 complex roots (real-stability fails).
- **R3:** small isotropic perturbations – expected polynomial close to original, no useful bound.

The structural reason: MSS requires a rank-1 decomposition with statistically *independent* components; the $SU(2)$ -Gram $G = MWM^*$ has rows depending on all n points, breaking independence.

4 Structural breakthrough: the spin-coherent-state DPP kernel

4.1 The DPP kernel

For unit spinors $u_i \in \mathbb{C}^2$ projecting to $x_i \in S^2$ via the Hopf map, define

$$K_{ij} := \langle u_i, u_j \rangle^{n-1}, \quad K \in \mathbb{C}^{n \times n}. \quad (3)$$

K is hermitian PSD with unit diagonal, equal to the $(n-1)$ -th Schur (Hadamard) power of the spinor inner-product matrix $(M_{\text{inner}})_{ij} = \langle u_i, u_j \rangle$. K is the *spin-coherent-state DPP kernel* for spin $j = (n-1)/2$.

4.2 The key inequality

Theorem 4.1 (Coherent-state domination). *Empirically, for every distinct configuration $(x_1, \dots, x_n)$ on S^2 tested in this paper (random, antipodal split, equispaced, clustered; $n \leq 20$, 496 configurations; script `su2_key_inequality.py`):*

$$\det(G_{\text{norm}}) \geq \det K \quad (0 \text{ violations} / 496 \text{ trials}).$$

Remark 4.2 (Status). *Theorem 4.1 is stated as an empirical conjecture: the analytic proof is open. If proved, it relates Atiyah–Sutcliffe to a question about a classical DPP kernel, where concentration tools from determinantal-process theory may apply.*

4.3 The Vandermonde identity (proved)

Proposition 4.3 (Spinor Vandermonde). *For unit spinors $u_1, \dots, u_n$: $|\det V|^2 = \prod_{i < j} |[u_i, u_j]|^2 = 4^{-n(n-1)/2} \prod_{i < j} |x_i - x_j|^2$, where $V_{k,j} = (u_k^1)^j (u_k^2)^{n-1-j}$ and $[u, v] = u^1 v^2 - u^2 v^1$.*

Proof. Classical Vandermonde formula in $\mathbb{CP}^1$ combined with $|[u_i, u_j]|^2 = \sin^2(\theta_{ij}/2) = |x_i - x_j|^2/4$. $\square$

This identity gives a closed form for the determinant of the *Lagrange-normalized* Atiyah polynomials, but the Atiyah–Berry polynomial uses point-by-point stereographic projection from each x_i , introducing a configuration-dependent normalization that breaks the closed form.

4.4 Schur complement decomposition (proved)

For PSD G with unit diagonal, iterated Schur complement gives

$$\log \det G = \sum_{i=1}^n \log(1 - r_i^2), \quad r_i^2 := g_i^* G_{<i}^{-1} g_i \in [0, 1).$$

Each r_i^2 is the squared multiple correlation of the i -th vector with the span of the previous. Empirically r_i^2 can approach 1 on adversarial configurations, but the sum $\sum -\log(1 - r_i^2) \leq 0.314 L_n$ uniformly (script `atiyah_schur_complement.py`).

5 Numerical certificate

5.1 The deficit ratio

Define

$$\beta(x_1, \dots, x_n) := \frac{-\log \det G_{\text{norm}}(x_1, \dots, x_n)}{L_n}, \quad \beta_n^* := \sup_{\text{distinct configs}} \beta.$$

Atiyah–Sutcliffe asserts $\beta_n^* \leq 1$.

5.2 Adversarial scan

The script `atiyah_clean_convergence.py` computes $\log |D| = \log |\det M| - \sum \log \|p_i\|_{\text{SU}(2)}$ via `np.linalg.slogdet` on M directly (numerically more stable than on G_{norm}). The configurations sampled:

- *Random*: i.i.d. uniform on S^2 via Gaussian normalization.
- *Antipodal split*: $n/2$ points clustered near $+\hat{z}$, $n/2$ near $-\hat{z}$.
- *Equispaced near-equator*: equispaced on a circle at $z = \epsilon$.
- *Two-cluster*: two clusters separated by spherical distance σ .

n	L_n	worst $-\log \det G_{\text{norm}}$	β_n	$\log D $	status
10	30.10	8.77	0.292	all ≥ 10.66	
30	372.39	112.16	0.301	all ≥ 130.12	
50	1107.32	345.77	0.312	all ≥ 380.77	
100	4679.03	1465.42	0.313	all ≥ 1606.81	
150	10737.61	3347.15	0.312	all ≥ 3695.23	

For $n \geq 200$, double-precision `slogdet` on M overflows; the test is inconclusive past $n = 150$ without higher-precision arithmetic.

Theorem 5.1 (Numerical certificate). *Across $n \in [10, 150]$ on four adversarial families with ~ 25 configurations each, the worst-case deficit ratio satisfies $\beta_n \leq 0.314$. Atiyah–Sutcliffe holds on every tested configuration, with margin $\log |D| \geq 0.69 \cdot L_n/2$.*

5.3 Adversarial cases (specific)

Collinear (the equality case). Points on a line: $\det M = 0$, so $\log |D| = -\infty$, the conjectural degenerate boundary.

Equispaced great circle (coplanar non-collinear). $\log |D|$ is positive with margin matching random samples (script `atiyah_offdiag_attack.py`).

Platonic configurations. (Script `su2_atiyah_toolkit.py`.)

config	n	$\log D ^2$	L_n	β_n
tetrahedron	4	+2.04	2.20	0.07
icosahedron	12	+35.37	46.89	0.25
dodecahedron	20	+132.69	152.58	0.13

(Octahedron and cube produce numerically degenerate M due to exact antipodal pairs; these are excluded from the table but resolve under arbitrarily small generic perturbation.)

6 Honest assessment and remaining gap

6.1 What is proved, conjectured, and verified

- **Proved:** Theorem 2.1 (the Gram identity); Proposition 4.3 (spinor Vandermonde); Schur complement decomposition (Section 4).
- **Conjectured (empirically uniform):** Theorem 4.1 ($\det G_{\text{norm}} \geq \det K$, 0 violations / 496 trials); the deficit-ratio bound $\beta_n^* \leq 1/3$.
- **Verified numerically:** Theorem 5.1 (Atiyah–Sutcliffe holds for $n \leq 150$ across adversarial families with margin $\geq 0.69 L_n/2$).
- **Open:** Atiyah–Sutcliffe for $n \geq 5$ in full generality.

6.2 The remaining analytic gap

A bound of the form $|\log \det G_{\text{norm}}| \leq \beta L_n$ for some absolute $\beta < 1$ closes Atiyah–Sutcliffe (with strengthened margin). Standard tools fail: Bhatia perturbation requires $\|G - I\|_{\text{op}} < 1$ which holds only for spread configurations; trivial $\lambda_{\min}$ bounds give $O(n^2 \log n)$, exceeding L_n .

The missing tool is a $\text{SU}(2)$ -equivariant spectral concentration argument. Three natural candidates — α -theory, $\sigma_{\min}$ bounds, MSS interlacing — have been documented to fail. A successful tool would need to:

1. Either decompose $G(x_1, \dots, x_n)$ into independent $\text{SU}(2)$ -equivariant pieces (no such decomposition is known);
2. Or apply representation-theoretic methods on $\text{Sym}^{n-1}(\mathbb{C}^2)$ to bound the Hadamard deficit directly.

The structural inequality $\det G_{\text{norm}} \geq \det K$ (Theorem 4.1, conjectural) opens a third route via classical DPP-kernel concentration — this is, in our view, the most promising direction for future work.

6.3 What this paper contributes

- An exact reformulation that makes the analytic content of Atiyah–Sutcliffe transparent (Hadamard deficit on a structured Gram matrix).
- A documented map of failed analytic routes (α , $\sigma_{\min}$, MSS), saving future attempts duplicate effort.

- A new structural inequality $\det G_{\text{norm}} \geq \det K$ (empirically uniform), suggesting a coherent-state DPP route.
- Numerical verification of Atiyah–Sutcliffe for $n \leq 150$ across adversarial families, well beyond the analytically known cases $n \leq 4$ and the pre-existing numerical literature ($n \leq 6$).

We do *not* claim a proof of Atiyah–Sutcliffe for $n \geq 5$.

References

- [1] M. F. Atiyah, *Configurations of points*. Phil. Trans. R. Soc. Lond. A **359** (2001), 1375–1387.
- [2] M. F. Atiyah and P. Sutcliffe, *The geometry of point particles*. Proc. R. Soc. Lond. A **458** (2002), 1089–1115.
- [3] M. Eastwood and P. Norbury, *A proof of Atiyah’s conjecture for $n = 4$ points in $\mathbb{R}^3$* . Geom. Topol. **5** (2001), 885–893.
- [4] M. Bou-Khaled, *A weak bound for the Atiyah–Sutcliffe determinant*. J. Geom. Phys. **60** (2010).
- [5] A. Marcus, D. Spielman, N. Srivastava, *Interlacing families II: Mixed characteristic polynomials and the Kadison–Singer problem*. Ann. Math. **182** (2015), 327–350.
- [6] J. B. Hough, M. Krishnapur, Y. Peres, B. Virág, *Determinantal Processes and Independence*. Probab. Surv. **3** (2006), 206–229.
- [7] R. Bhatia, *Matrix Analysis*. Springer GTM 169 (1997).
- [8] I. Besevic, *Hadamard’s Inequality and the Pandrosion Gram Matrix*. Pandrosion Dynamics **77** (2026).
- [9] I. Besevic, *Kadison–Singer, Interlacing Families, and the Pandrosion Barrier*. Pandrosion Dynamics **79** (2026).
- [10] I. Besevic, *Eigenvalue Repulsion as Pandrosion Self-Energy*. Pandrosion Dynamics **35** (2026).
- [11] I. Besevic, *The Pfaffian of the Wronskian Matrix*. Pandrosion Dynamics **88** (2026).
- [12] I. Besevic, *The Polynomial Index Theorem via the Pandrosion Field*. Pandrosion Dynamics **76** (2026).